Siddharth Kalra

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Education

Delhi Technological University **2023 – 2027**
B.Tech in Mathematics and Computing, ***CGPA: 7.1/10.0*** *New Delhi, India*

Sant Gyaneshwar Public School (CBSE) **2023**
Higher Secondary Education, ***Grade: 92%*** *New Delhi, India*

Experience

WINK **Jun 2025 – Jul 2025**
Software Engineering Intern *Gurgaon, India*

- Architected secure, stateless authentication workflows using **JSON Web Tokens (JWTs)**, replacing session-based auth to improve scalability across services.
- Integrated **"Sign in with Google"** via **OAuth 2.0**, reducing signup friction and enabling one-click login for end users.
- Configured and deployed **Keycloak** for **Single Sign-On (SSO)**, centralizing access control across **multiple web applications** on the platform.

Projects

Housing Price Prediction | *Python, Random Forest, Scikit-Learn, Pandas, TensorFlow/Keras* |

- Derived **12 domain-specific features** (TotalSF, QualxArea, HouseAge, etc.) from **79 raw columns**, with **5 features ranking in the top-20** by importance score.
- Optimized a **Random Forest** regressor via **RandomizedSearchCV** (**100 iterations**, **5-fold CV**), improving RMSLE from **0.1436 to 0.1321** — a **4.7% gain** over baseline.
- Benchmarked against a **DNN baseline** (256→128→64→32→1) with ensemble averaging; the RF model outperformed the neural network using hand-crafted features alone.

CIFAR-10 Image Classifier | *Python, TensorFlow/Keras, Scikit-Learn* |

- Designed a **3-block CNN** (TensorFlow/Keras) with BatchNorm, Dropout, and data augmentation, achieving **83.13% validation accuracy** across **10 image categories**.
- Optimized hyperparameters (filters, dropout rate, learning rate) via **RandomizedSearchCV** with **3-fold cross-validation**, improving model generalization.
- Automated training using **EarlyStopping**, **ReduceLROnPlateau**, and **ModelCheckpoint** callbacks, cutting manual tuning effort and training time.

Real-Time Calculator (OCR) | *Python, PyTorch, OpenCV, CNN* |

- Built a real-time calculator that detects handwritten digits and evaluates arithmetic expressions directly from images.
- Trained a **CNN** (**PyTorch**, **OpenCV**) on an augmented digit dataset to improve robustness across varied handwriting styles.
- Achieved **97.5% accuracy** on prefix-notation and **93% accuracy** on postfix-notation expression evaluation.

Technical Skills

Programming Languages: C, C++, Python
Web Technologies: HTML, CSS, Bootstrap, JavaScript, Node.js, Express.js, EJS, SQL, MongoDB
ML / Data Science: TensorFlow, Keras, PyTorch, OpenCV, Scikit-Learn, Pandas, NumPy, Matplotlib
Tools: Git, GitHub
Coursework: Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks

Achievements & Extra-Curricular Activities

- **JEE Main:** Secured **98.26 percentile** (AIR 20,175) out of **13 lakh candidates**.
- **Competitive Programming:** Solved **300+ DSA problems** across competitive programming platforms.
- **Badminton:** Secured second place in Badminton Zonals and represented Apeejay School, Pitampura at the Interzonal level (2018).
- **Chess:** Won third position in the Under-17 Chess Competition organized by the Uttarakhand Sports Association (2021); active competitive chess player with a Chess.com Rapid rating of ∼**1500**.